



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

ASSIGNMENT 2

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1) Write a program to implement

CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Sample{ 4 int x; 5 public: 6 void set(int x){ 7 this->x=x;} 8 void show(){ 9 cout<<x;}}; 10 int main(){ 11 Sample obj; 12 Sample *p=&obj; 13 obj.set(10); 14 obj.show(); 15 cout<<endl; 16 p->set(20); 17 p->show(); 18 return 0;}</pre>	<pre>10 20 === Code Execution Successful ===</pre>

2) Write a program to swap private values of two classes using a friend function.

CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class B; 4 class A{ 5 int x; 6 public: 7 A(int a){x=a;} 8 friend void swap(A&,B&); 9 void show(){cout<<x;}; 10 class B{ 11 int y; 12 public: 13 B(int b){y=b;} 14 friend void swap(A&,B&); 15 void show(){cout<<y;}; 16 void swap(A &a,B &b){ 17 int temp=a.x; 18 a.x=b.y; 19 b.y=temp;} 20 int main(){ 21 A a(5); 22 B b(10); 23 swap(a,b); 24 a.show(); 25 cout<<endl; 26 b.show(); 27 return 0;}</pre>	<pre>10 5 === Code Execution Successful ===</pre>

3)Write a program to add data objects of two different classes using friend functions.

CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class B; 4 class A{ 5 int x; 6 public: 7 A(int a){x=a;} 8 friend int add(A,B);}; 9 class B{ 10 int y; 11 public: 12 B(int b){y=b;} 13 friend int add(A,B);}; 14 int add(A a,B b){ 15 return a.x+b.y;} 16 int main(){ 17 A a(10); 18 B b(20); 19 cout<<add(a,b); 20 return 0;}</pre>	<pre>30 === Code Execution Successful ===</pre>

4) Write a program to demonstrate the working of a friend class.

CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class A{ 4 int x; 5 public: 6 A(){x=100;} 7 friend class B;}; 8 class B{ 9 public: 10 void show(A a){ 11 cout<<a.x;}; 12 int main(){ 13 A a; 14 B b; 15 b.show(a); 16 return 0;}</pre>	<pre>100 === Code Execution Successful ===</pre>

5) Write a program using an array of objects to display the area of multiple rectangles.

CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class Rectangle{ 4 int length,breadth; 5 public: 6 void get(){ 7 cout<<"Enter length and breadth: "; 8 cin>>length>>breadth;} 9 void area(){ 10 cout<<"Area = "<<length*breadth<<endl;}; 11 int main(){ 12 int n; 13 cout<<"Enter number of rectangles: "; 14 cin>>n; 15 Rectangle r[n]; 16 for(int i=0;i<n;i++) 17 r[i].get(); 18 for(int i=0;i<n;i++) 19 r[i].area(); 20 return 0;}</pre>	<pre>Enter number of rectangles: 3 Enter length and breadth: 2 7 Enter length and breadth: 3 8 Enter length and breadth: 4 9 Area = 14 Area = 24 Area = 36 === Code Execution Successful ===</pre>

6) Write a program to define function `cube()` as an inline function for calculating the cube of a number.

CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 inline int cube(int x){ 4 return x*x*x; 5 } 6 int main(){ 7 int n; 8 cin>>n; 9 cout<<"Cube= "<<cube(n); 10 return 0; 11 }</pre>	<pre>12 Cube= 1728 === Code Execution Successful ===</pre>

7) Write a program to pass an object as an argument and return the object from a function:

A) CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class A{ 4 public: 5 int x; 6 }; 7 A change(A a){ 8 a.x=50; 9 return a; 10 } 11 int main(){ 12 A a; 13 a.x=10; 14 a=change(a); 15 cout<<a.x; 16 return 0; 17 }</pre>	<pre>50 === Code Execution Successful ===</pre>

B) CODE & OUTPUT:

main.cpp	Output
<pre>1 #include <iostream> 2 using namespace std; 3 class A{ 4 public: 5 int x; 6 void change(A *a){ 7 a->x=50; 8 } 9 int main(){ 10 A a; 11 a.x=10; 12 change(&a); 13 cout<<a.x; 14 return 0; 15 }</pre>	<pre>50 === Code Execution Successful ===</pre>